FACULTY OF COMPUTING

UNIVERSITI TEKNOLOGI MALAYSIA

NETWORK COMMUNICATIONS

SECR1213-12

SEMESTER 1 2024/2025

PROJECT TASK 2: INITIAL DESIGN - PRELIMINARY ANALYSIS

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GROUP 5

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Questions

Category 1: Network Connectivity

1. Which network cable types are most appropriate for the computers being connected, and why was it chosen?

   The most suitable network cable for connecting computers is the Cat6 Ethernet Cable, a LAN connection that complies with the Ethernet protocol and is made particularly for network data transmission. It can carry data at up to 10 Gbps over a maximum distance of 55 meters, which makes it appropriate for high-speed connections in faculty. Performance is steady and consistent because of its twisted pair design, which reduces interference. Cat6 is adaptable and perfect for managing various devices and applications, including online learning, sharing data, and video conferencing. It is an appropriate choice for basic faculty usage because it is more affordable than fiber optics. It also enables larger bandwidths, enabling network upgrades, scalability, and future-proofing.

2. What type of connectivity is preferred for the labs (wired, wireless, or both)?

   To meet their individual needs, every room has both wired and wireless connectivity. There is a webcam in the video conferencing room. While wired connections provide steady audio and video during meetings, wireless connectivity enables other devices to join easily if necessary. In order to provide flexibility, the Student Lounge allows students to bring their own computers and connect wirelessly for convenience. For stationary settings that require more reliable connections, wire solutions are also available. The Hybrid Classroom and General Lab have PC and projector.Wired connections guarantee that these fixed equipment run reliably throughout lectures and presentations, while wireless connections allow for the use of other devices by teachers and students, such as laptops and tablets.The Cisco Network Lab, which offers wireless flexibility but depends on cable connections for stability, has a smartboard, camera, PCs, and a server for demonstration.Similar to this, the Embedded Lab uses wifi for extra device connectivity and wired connections for essential devices including its smartboard, PCs, and projector. In any environment, this combination guarantees reliability, flexibility and ease.

3. What wireless standard is required or preferred for the faculty network?

   For a Faculty network, Wi-Fi 6 is preferred for a faculty network compared to Wi-Fi 5 (802.11ac), it offers a number of significant modifications, such as faster speeds, more efficiency, greater capacity, and better performance in high-density settings. Wi-Fi 6 is perfect for university networks that may have numerous users and devices connected at once because it is made to support more devices at once. Furthermore, Wi-Fi 6 offers enhanced security capabilities (such WPA3), reduced latency, and greater range, all of which increase network dependability and user experience overall.

4. What type of cabling is recommended for the network backbone and workstations?

   Fiber-optic cabling is recommended for the network backbone due to its high speed, low latency, and ability to span great distances without signal loss. Fiber-optic connections, such as single-mode fiber, can perform bandwidth-intensive operations such as connecting multiple labs, a hybrid classroom, and a video conference room, ensuring dependable and fast data transfer. For workstation connections, Cat6 or Cat6a twisted-pair cables are recommended. These connections enable up to 10 Gbps over short distances and are cost-effective for connecting devices such as computers, printers, and IoT peripherals in the lab. In addition, Cat6a offers improved shielding to prevent crosstalk and interference, which is useful in high-density areas such as labs.

5. What types of network devices are suitable for an academic institution of this size?

   For an academic institution of this size, enterprise-grade routers and switches are essential for a reliable and scalable network backbone, supporting features like VLANs and traffic prioritization. Wireless access points with Wi-Fi 6 technology ensure high-speed connectivity and efficiency in areas with dense user traffic, such as the student lounge and hybrid classroom. Network security appliances, including firewalls and intrusion detection systems, protect against cyber threats and ensure secure operations. Specialized labs require tailored devices: the Cisco Network Lab needs educational routers and switches for hands-on learning, while the Embedded Lab benefits from IoT tools like microcontroller kits and sensors. High-speed cabling, such as fiber optic for the backbone and Cat 6 for internal connections, along with appropriate power and cooling systems, ensures seamless and efficient network performance.

6. What are the types of routers that are suitable to be used in each lab and in the building?

Based on research, the common types of routers are wireless router, modem router, brouter, access router, edge router and distribution router.

For General Lab, the suitable router type is access router. This is because general labs normally only require moderate bandwidth for web browsing and access to educational resources. Since an access router is designed to provide basic LAN and WAN connectivity for small-to-medium environments, it is more suitable to be used in the General Lab.

Nevertheless, the edge router is more suitable to be used in Cisco Network Lab. In general, Cisco Network Lab is used for students to learn on network simulations tools by using virtual or real workstations and servers. Based on research, edge routers will connect multiple networks and send packets across them. They also provide some protocol support such as Border Gateway Protocol (BGP) for connectivity. By concluding this information, we found that edge routers can manage data traffic that enters and exits the network easily, which is actually suitable for use in Cisco network Lab.

Nonetheless, Embedded Labs, which work with IOT devices, sensors and digital are required bridging and routing capabilities to ensure hybrid connectivity through wired and wireless devices. Brouter is most suitable to be used in this lab since it can serve as both router and bridge depending on the traffic, which support the bridging and routing capabilities.

Lastly, Wireless Routers are suitable to use in Hybrid Classroom, Academic Office and Student Lounge. This is because in these areas, users normally will bring their own laptops and wireless networks are more comfortable for them to connect to WIFI easily.

Other than that, wireless routers provide stable and high speed connectivity, which is actually suitable for these areas.

7. What is the difference between switch and router? Where should they be placed in this building?

   Switch is used for connecting devices in the same network. Thus, it is suitable to be used for the purpose of communication between network devices, such as printers and computers in the same network.

   Router is used to connect multiple networks and send packets between networks. For example, a router is used to interconnect with different labs and different facilities inside the campus.

   Switch should be placed in all of the areas in the building, such as at least one switch in Cisco Network Lab, Embedded Lab, General Lab, Hybrid Classroom, Student Lounge, and Academic Office. It will connect to almost all network devices, such as computers, smartboard, TVs and so on. Router should be placed in the server room and connected with the switch in all areas, to provide network access to the school network and other networks.

Category 2: Performance and Bandwidth

1. What requirements must be met for high-speed internet connections in order to satisfy computing demands in labs, classrooms, and video conferencing rooms?

   A variety of requirements must be fulfilled in order to provide high-speed internet in labs, classrooms, and video conferencing rooms. It is crucial to have a bandwidth management system with dynamic bandwidth allocation. By providing bandwidth based on user requirements, this system optimizes network capacity while ensuring that users have access to the resources they desire. Additionally, it can allocate specific bandwidth and time slots, which is important in areas with a high user density. These technologies control the bandwidth that ISPs give, guaranteeing a steady internet connection speed for activities like high data utilization in labs and real-time communication. In order to avoid disruptions, Quality of Service (QoS) also makes sure that high-priority applications, like video conferencing, receive the bandwidth they require. High-bandwidth and low-latency connections are necessary for video conferencing rooms to ensure smooth communication

2. What is the maximum number of devices expected to connect to the wireless network at any given time in the building, especially in common areas like the student lounge?

   When using Wi-Fi 6, the network can support a higher number of devices at once compared to older Wi-Fi standards. According to the size of the area and the number of users, the maximum number of devices that can connect at any given moment in common spaces like the student lounge could be somewhere between 500 and 1,500. Wi-Fi 6 is perfect for locations where a large number of employees or students are connecting at once since it can handle many devices without slowing down. Even with numerous devices online, this guarantees a quicker and more reliable connection

3. What is the required bandwidth and speed for each lab and the student lounge?

   Each lab requires a minimum bandwidth of 1 Gbps to support high-speed internet for Embedded lab, Cisco Network lab, and resource-intensive tasks like network simulations and real-time applications. The student lounge requires at least 500 Mbps, supported by Wi-Fi 6 (IEEE 802.11ax), to enable smooth connectivity for multiple simultaneous users engaged in casual browsing, online collaboration, and media streaming.

   These specifications align with the Fourth Industrial Revolution (4IR) educational standards, emphasizing the need for advanced digital tools in learning environments. High-speed fiber-optic cabling and managed switches will ensure reliable, scalable, and efficient network performance, preparing the infrastructure for future growth and increased user demands.

4. How does Quality of Service (QoS) ensure optimal performance for critical applications in each lab of the project?

In our project, where each lab has distinct duties that demand varying bandwidth, low latency, and minimal packet loss, QoS is crucial for sustaining the performance of various applications in a network. Real-time data processing and high-priority traffic handling were necessary for the Cisco Networking Lab, therefore QoS will allot dedicated bandwidth to the simulation tools in order to maintain steady performance even during periods of high demand. To avoid delays, traffic from the simulation server will be given a higher priority than traffic from regular surfing. Since the Embedded Lab involves constant data interchange between sensors and processing devices, QoS makes guarantees that commands and data are sent with minimal latency. In contrast, QoS gives video conferencing traffic during online lectures a higher priority than overall web usage for the General Purpose Lab. The purpose of this action is to ensure that students have a seamless experience when working on time-sensitive tasks.

Category 3: Scalability and Future-Proofing

1. What are scalability requirements for the network, considering the 15% growth in users over the next four years?

   Scalability guarantees that the network can expand in response to rising demand. As the number of users and devices increases, adding ports is made simple by modular network designs, such as modular switches. Because they guarantee that the network can grow without interfering with currently offered services, these switches are adaptable. Additionally, VLANs offer logical device grouping, which lowers network congestion and boosts productivity. Adopting Wi-Fi 6 might be taken into consideration as the number of wireless devices rises. It can manage more devices at once, offers faster speed, greater efficiency, and the ability to sustain excellent performance even as user density increases. Finally, IPv6 adoption and the use of higher-capacity cabling, such as Cat6a, will be required to accommodate more devices. Enough IP addresses are guaranteed by IPv6 to support the growing number of connected devices.

Category 4: Security

1. What security measures should be implemented to protect sensitive faculty data from breaches?

Both network security and user education are necessary to protect sensitive data. Installing firewalls first will filter traffic and prevent the majority of malicious material from entering or exiting. Advanced encryption systems, such as WPA3, will jumble the data for the wireless connection element, ensuring that only authorized users can read it. Second, users can operate remotely and maintain the privacy of their connections by using Virtual Private Networks (VPNs), which offer a secure tunnel. Third, Multi-Factor Authentication (MFA) increases security by requiring users to provide a one-time password (OTP) to their phone in order to confirm their identity. Fourth, instructors can update the software on a regular basis to stop hackers from taking advantage of known flaws. Fifth, in order to reduce human error, training programs that teach employees and students how to spot phishing attacks and steer clear of fraudulent websites are essential.

2. How can network infrastructure support remote access for academic staff and students?

   Secure connections such as Virtual Private Networks (VPNs), which build encrypted tunnels to guarantee data secrecy while transmission over public networks, facilitate remote access. Depending on the faculty's budget, VPN deployment might range from low-end software-based systems to high-availability hardware. Additionally, to lessen the likelihood of data breaches, Role-Based Access Control (RBAC) ensures that users only access information pertinent to their responsibilities. By enabling real-time communication and document sharing, platforms such as Google Workspace can further streamline collaboration and boost productivity for both distant educators and learners.

3. How secure access will be provided to external contractors or non university students if required?

   Secure access to school network resources for guests can be solved in multiple ways. First and foremost, setting up a guest network can solve this problem. By setting up an external wireless network for guests, this can ensure schools have the full control on which network resources that guest can access to prevent guest access on sensitive data. Additionally, temporary access accounts are also one of the ways to solve this issue. This can be done by creating time-limited accounts that also restrict access to sensitive information. This way is actually more easy to set up and affordable compared to setting up a guest network, but it will waste more time and resources in the long-term since it requires creating and deleting accounts multiple times.

Category 5: Management and Monitoring

1. What are the management and monitoring requirements for the network?

   Effective network management and monitoring ensure optimal performance, reliability, and security. Implementing tools like SolarWinds or Cisco DNA Center enables real-time monitoring to detect issues such as bottlenecks and anomalies, ensuring swift resolutions. Configuration management streamlines device setup, ensures consistency, and supports backups for quick restoration in case of failures. Automated alerts and troubleshooting features allow proactive issue identification and resolution, minimizing downtime.

   Performance analytics provide insights into traffic patterns and usage, helping optimize resources and maintain consistent performance. Remote management capabilities further enhance efficiency by allowing centralized control of distributed networks, reducing the need for on-site interventions. These functionalities together ensure the network remains robust, scalable, and aligned with the institution's growing needs.

Category 6: Redundancy and Reliability

1. How should power backup and redundancy be addressed?

   Power backup and redundancy techniques are critical for maintaining network operations during power outages or breakdowns. Install Uninterruptible Power Supplies (UPS) in key devices such as routers, switches, and servers to offer short-term power and surge protection. To increase reliability, use devices with dual redundant power supplies that are connected to separate power sources. Install backup generators with Automatic Transfer Switches (ATS) for long-term outages to ensure uninterrupted power transfer. Install sophisticated Power Distribution Units (PDUs) to monitor and balance power distribution. Regular testing and maintenance of UPS batteries, generators, and power connections is critical for avoiding failures. Furthermore, environmental monitoring systems can track electricity usage and conditions, notifying administrators of potential problems.

Category 7: Lab-Specific and Custom Features

1. What type of network design and configuration is ideal for each lab to meet its unique requirements?

   For Cisco Network Lab, a hierarchical network architecture comprising core, distribution, and access layers is advised. While access switches link end-user devices, such as workstations, core switches guarantee fast data transport. For best results, virtual local area networks (VLANs) should be set up to keep administration and schooling apart. Low-latency communication should be given top priority in the Embedded Lab network in order to facilitate IoT experimentation. It is best to use a star topology to link IoT gateways, sensors, and controllers to a central switch. By employing VLANs for network segmentation, experiment data and regular traffic are kept apart. Simple star topology, in which every desktop computer is linked to a central switch for convenient management and effective resource use, is also adequate for general purpose labs. In order to facilitate the flexible use of laptops and other devices, Wi-Fi 6 access points can be used in addition to cable connections.

2. What are the pros and cons for including a server room inside this building?

   Server room is a room where enterprise servers are kept. It will require a controlled environment such as temperature, humidity, electric power and so on. It is also known as a “data center” that stores its own enterprise data inside it to ensure full control and security on the data.

   There are several benefits to including a server room. First, network performance will be improved since local servers can reduce the latency for communication and access between network devices, and ensure faster data transfer in the building. Data security also will be improved since all data is stored in a local server. Other than that, universities also can customize the hardware and software configurations to meet the needs of specific rooms in the building, which can make the network distribution more equal and efficient in each room.

However, the server room also has its disadvantages. Server rooms will require a high initial and maintenance costs, since servers, cooling systems, security and so on are expensive. Additionally, hiring some high expertise staff will also increase the cost, which is actually expensive. Nevertheless, the server room will occupy some spaces in the building, which originally can be used for other areas such as the lab or study area.

3. There are a lot of network devices such as PCs, laptops, projector, smartboard, microphone, Server (Put in Cisco Network Lab for learning purpose), TVs and so on. Which of them need to connect with the server in the server room (wired or wireless), and which of them is not needed?

   Devices such as PCs, laptops, smartboards, IoT equipment, WIFI projector, servers in Cisco Lab, and TVs need to connect with servers in the server room. This is because they need to have access to the server to get the data and do their own work. For example, WIFI projector will require connection with the server to get the resources from the server. In these network devices, normally only laptops will be wireless, and others will be wired.

   Nevertheless, devices such as standalone projector, personal devices and some simple lab equipment such as sensors might not require direct connection with the server in the server room. For standalone projector, normally it will be set up initially and can be easily used via HDMI or USB, thus it does not require network connection. Other than that, personal devices that do not access university information will not connect with the server in the server room. For standalone projector, it will be wired and personal devices will be wireless.

Feasibility Assessment

The proposed network architecture for the Faculty of Computing (FC) intends to meet the infrastructure requirements of a new two-story building that would include labs, a hybrid classroom, a video conferencing room and a student lounge. This project aims to create a dependable, scalable, and secure network that meets current technology requirements while planning for future expansion. The project's feasibility is assessed based on technical, operational, and economic aspects.

Technically, the project is feasible due to the availability of scalable and high-performance networking solutions.The design includes modular switches, VLANs, and Cat6 cable to handle the current user base while accommodating a 15% increase in students and staff over the next four years. Secure technologies such as firewalls and intrusion detection systems improve network security, while high-speed connections of at least 1 Gbps satisfy the requirements for modern, 4IR-aligned facilities. The infrastructure also supports future wireless network extension, ensuring a future-proof design.

Operationally, the design encourages efficiency and dependability with centralized network management tools like Cisco Packet Tracer. This strategy makes monitoring and troubleshooting easier, lowering the workload on IT professionals. The phased implementation of the infrastructure reduces disruptions, allowing for seamless integration without interfering with ongoing activities. Furthermore, the network's scalability allows for seamless updates and extensions, preserving the institution's operational stability and readiness for growth.

Economically, the project is well within its allocated budget of RM 5 million. The estimation of costs indicate that RM 2.5 million will be spent on core networking devices, RM 1 million on workstations, and RM 1 million on cabling and installation, with RM 0.5 million set aside for eventualities and miscellaneous needs. By using high-quality yet low-cost equipment, the project reduces long-term maintenance expenses while providing a strong solution. Hence, the project is economically viable, as it balances urgent needs with long-term provisions.

References

1. Abbey, N. (2023, April 1). *What are Routers in Networking?* STL Tech. https://stl.tech/blog/what-are-routers-in-networking/
2. Ascendant Technologies, Inc. (2024, February 21). Network management: Definition, importance, and best practices. Ascendant Technologies. Retrieved from https://ascendantusa.com/2024/02/21/network-management-guide/
3. Bernaldo, A. (2024, August 16). *What is a LAN Cable? Understanding Types, Uses, & Differences.* GEARit. https://www.gearit.com/blogs/news/what-is-a-lan-cable#:~:text=In%20summary%2C%20while%20both%20cables%20are%20essential%20for,various%20devices%20in%20both%20home%20and%20office%20settings.
4. *Cisco Modeling Labs - Enterprise V2.0.* (2024, November 21). [Video]. Cisco. https://www.cisco.com/site/us/en/learn/training-certifications/training/modeling-labs.html#:~:text=Cisco%20Modeling%20Labs%20is%20an,designing%2C%20testing%2C%20and%20troubleshooting
5. Gager, R. (2013). Keys to backup power system success: Features, flexibility, and redundancy. FacilitiesNet. Retrieved from https://www.facilitiesnet.com/powercommunication/article/Keys-To-Backup-Power-System-Success-Features-Flexibility-And-Redundancy--14802.
6. Gentile, A. F., Fazio, P., & Miceli, G. (2021). A Survey on the Implementation and Management of Secure Virtual Private Networks (VPNs) and Virtual LANs (VLANs) in Static and Mobile Scenarios. Telecom, 2(4), 430–445. https://doi.org/10.3390/telecom2040025
7. Ghoshal MKrishna SGringoli FWidmer JKoutsonikolas D(2024)A First Look at 160 MHz WiFi 6/6E in Action: Performance and Interference Characterization2024 IFIP Networking Conference (IFIP Networking)10.23919/IFIPNetworking62109.2024.10619856(489-495)Online publication date: 3-Jun-2024 https://doi.org/10.23919/IFIPNetworking62109.2024.10619856
8. J. P. Agrawalkand and Y. E. Cherner, "A classroom/distance learning engineering course on optical networking with virtual lab," 2008 IEEE Region 8 International Conference on Computational Technologies in Electrical and Electronics Engineering,

Novosibirsk, Russia, 2008, pp. 73-77, doi: 10.1109/SIBIRCON.2008.4602554. keywords: {Region 8;Distance learning}, https://ieeexplore.ieee.org/abstract/document/4602554

9. M. S. Afaqui, E. Garcia-Villegas and E. Lopez-Aguilera, "IEEE 802.11ax: Challenges and Requirements for Future High Efficiency WiFi," in IEEE Wireless Communications, vol. 24, no. 3, pp. 130-137, June 2017, doi: 10.1109/MWC.2016.1600089WC.keywords: {IEEE 802.11 Standard;Throughput;Wireless LAN;Interference;OFDM;Parity check codes}, https://ieeexplore.ieee.org/abstract/document/7792393
10. Mozaffariahrar, E., Theoleyre, F., & Menth, M. (2022). A Survey of Wi-Fi 6: Technologies, Advances, and Challenges. Future Internet, 14(10), 293. https://doi.org/10.3390/fi14100293
11. Network Drops. (n.d.). The different types of network cabling. Network Drops. Retrieved from https://networkdrops.com/blog/the-different-types-of-network-cabling/#:~:text=Fiber%20Optic%20Cable&text=Fiber%20optic%20cables%20have%20become,resistance%20to%20moisture%20and%20lighting.
12. Projector, W. P. (2023, March 21). Smart Projector vs WiFi Projector: Which one should you choose? *WoW 3D Displays.* https://www.wow3d.com.au/post/smart-projector-vs-wifi-projector-which-one-should-you-choose#:~:text=No%20need%20for%20any%20additional,speaker%20for%20high%2Dquality%20sound
13. Schwab, K. (2016). The Fourth Industrial Revolution: What it means and how to respond. World Economic Forum. Retrieved from https://www.weforum.org/stories/2016/01/the-fourth-industrial-revolution-what-it-means-and-how-to-respond/
14. Sharma, D. (2024, October 1). Network hardware devices explained - Hub, switch, router, firewall, etc. UniNets. https://www.uninets.com/blog/network-hardware-devices
15. *Small business IT explained in 60 seconds or less: Switches vs. Routers.* (2024, October 29). [Video]. Cisco. https://www.cisco.com/c/en/us/solutions/small-business/resource-center/networking/network-switch-vs-router.html

16. Srishti Priya Chaturvedi, Vidhu Baggan, & Kumar, P. (2020). Comparative Analysis of Traditional Virtual-LAN with Hybrid Software Defined Networking Enabled Network. Comparative Analysis of Traditional Virtual-LAN with Hybrid Software Defined Networking Enabled Network. https://doi.org/10.1109/cicn49253.2020.9242631
17. Tarn, J. M., Razi, M. A., Wiley, N., & Hsu, J. (2007). Exploring user perception of wireless campus. International Journal of Mobile Communications, 5(6), 710. https://doi.org/10.1504/ijmc.2007.014183
18. Von Hollen, E. (2024, November 13). Advantages and disadvantages of server room and data center. *ThomasTech.* https://www.thomastechllc.com/articles/advantages-and-disadvantages-of-server-room-and-data-center
19. *What is a Server Room? - GIGABYTE Global.* (n.d.). https://www.gigabyte.com/Glossary/server-room
20. Xu, Z., & Ni, J. (2020, December 1). Research on network security of VPN technology. https://doi.org/10.1109/ICISE51755.2020.00121

Meeting Minutes #1

<table>
<tr><td colspan="2">Date/Time</td><td colspan="2">27 Nov 2024 11:30am</td></tr>
<tr><td colspan="2">Location</td><td colspan="2">Online (Google Meet)</td></tr>
<tr><td colspan="2" rowspan="2">Agenda</td><td colspan="2">1. Discuss task 2 and choose method to complete task 2</td></tr>
<tr><td colspan="2">2. Determine number of preliminary analysis need to be done</td></tr>
<tr><td colspan="2">Meeting Host</td><td colspan="2">Joanne Ching Yin Xuan</td></tr>
<tr><td colspan="4">Attendance</td></tr>
<tr><td colspan="2">Name</td><td>Time</td><td>Reason For Absence</td></tr>
<tr><td colspan="2">Goh Jiale</td><td>11:30am</td><td>-</td></tr>
<tr><td colspan="2">Joanne Ching Yin Xuan</td><td>11:30am</td><td>-</td></tr>
<tr><td colspan="2">Evelyn Goh Yuan Qi</td><td>11:30am</td><td>-</td></tr>
<tr><td colspan="2">Lim Yu Han</td><td>11:30am</td><td>-</td></tr>
<tr><td colspan="4">Minutes</td></tr>
<tr><td>No.</td><td>Item Discussed</td><td>Details</td><td>Person-In-Charge</td></tr>
<tr><td>1</td><td>Brief about Task 2</td><td>Explain the requirement of the preliminary analysis</td><td>Joanne</td></tr>
<tr><td>2</td><td>Discuss Task 2</td><td>Research method and number of question need to be done which is 20 has been chosen to complete task 2</td><td>All</td></tr>
<tr><td rowspan="2">3</td><td rowspan="2">Distribute number of questions and answer that need to find by each members</td><td>Each members need to find out 5 questions and answers with reference</td><td>All</td></tr>
<tr><td>Organize all the questions and answers into different categories which is Network Connectivity, Performance and Bandwidth, Scalability and Future-Proofing, Security, Management and Monitoring, Redundancy and Reliability, Lab-Specific and Custom Features</td><td>Jiale</td></tr>
<tr><td>4</td><td>Feasibility</td><td>Feasibility of the project</td><td>Evelyn</td></tr>
<tr><td>5</td><td>Finalize Task 2</td><td>Check all the questions and answers</td><td>Evelyn & Jiale</td></tr>
</table>

6	Meeting ended	12:30pm	